Transport and Curvature Certificates for the Totient Constant

a Lean 4 companion to the Erdős #249 formalisation

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Abstract

Let $S = \sum_{n\geq 1} \varphi(n)/2^n$ and $R_N = \sum_{j\geq 1} \varphi(N+j)/2^j$. The irrationality of S is Erdős Problem 249 and remains open. We formalise exact secondary reductions built from finite differences of the tails R_N: curvature, exponent-transport and prime-jump commutators, and an anchored four-vertex operator at $H, 3H, 5H, 15H$. Each object splits into an integer dyadic window and a rigorously bounded tail, giving a decidable modular certificate for non-integrality. We prove that old Möbius divisor channels are affine and are annihilated by the relevant moments, isolate the new-channel migration at a prime jump, and show that every fixed-precision valuation–unit word admits a centred completion. The finite non-integrality criteria and no-go theorem are unconditional; every irrationality conclusion remains conditional on an explicitly named unbounded supply. No such supply, and hence no solution of Problem 249, is claimed.

1 Scope and logical boundary

The main exposition [3] develops the direct certificate reduction for S, proves an unconditional denominator exclusion, and records the open boundary. This companion studies finite-difference operators on the same tail sequence. They annihilate affine contributions from divisor channels already present at an lcm height, but the remaining new channels must still be shown to produce non-integral values at unboundedly many heights. Thus every endpoint below has the form

$$\text{unbounded supply of finite certificates} \Longrightarrow S \notin \mathbb{Q},$$

with the antecedent open. A certificate at one height is finite evidence, not evidence for the cofinal quantifier. The fixed-precision no-go theorem is a boundary on one local strategy, not a global impossibility result.

The Lambert-series background is classical. The open problem is recorded in the Erdős–Graham collection and maintained catalogue [1, 2]. We make no priority claim for finite differences or affine moment cancellation as general techniques. The contribution here is their precise integration into checked identities, finite non-integrality criteria, conditional hypotheses, and an obstruction theorem.

Small blue source marks link the displayed claims to their formal proofs and may be ignored on a first reading.

2 Dyadic windows

For a finite integer combination $X(H) = \sum_i c_i R_{m_i H}$, truncate each tail at depth L. Multiplication by 2^L gives

$$2^L X(H) = A(H, L) + E(H, L),$$

where $A(H,L) \in \mathbb{Z}$ is computable. If $|E(H,L)| \le B(H,L)$, then $X(H) \notin \mathbb{Z}$ whenever

$$B(H,L) < A(H,L) \bmod 2^L < 2^L - B(H,L).$$

The reusable soundness theorem is `notMem_int_of_window_remainder_bound` PriJumWin. Lean checks the exact decomposition, radius, and concrete modular arithmetic; an external analytic or arithmetic argument must provide unbounded instances.

3 Curvature on the lcm cone

Set

$$Q_H = R_{4H} - 3R_{2H} + 2R_H.$$

Its depth-L window $C(H,L)$ satisfies

$$2^L Q_H = C(H,L) + E_C(H,L), \qquad |E_C(H,L)| \le 6H + 3L + 3.$$

The exact split and bound are `two_pow_mul_curvature_eq_window_add_shifted` CurCar and `abs_curvatureShiftedTail_le` CurCar. The central residue criterion therefore proves $Q_H \notin \mathbb{Z}$ by `curvature_notMem_int_of_sharpCurvatureCert` CurCar. An unbounded supply implies irrationality through `irrational_totientSeries_of_sharpCurvatureSupply` CurCar, but that supply is not proved.

The window evolves by $C(H,L+1) = 2C(H,L) + d(H,L+1)$, checked in `curvatureWindow_succ` CurCar. Exact phase-certificate and common-carrier formulas expose possible routes to a supply theorem without asserting one.

4 Exponent transport and affine cancellation

If $p \mid H$, multiplication by p adds no squarefree divisor support:

$$d \mid pH \iff d \mid H \qquad (d \text{ squarefree}).$$

This is `squarefree_dvd_mul_iff_of_dvd` ExpOnlTra. The exact residue kernel is affine, not merely periodic:

$$K_d(N) = N\, a_d(N) + b_d(N),$$

where both a_d and b_d are periodic modulo d. The decomposition is `transportResidueKernel_eq_affineState` ExpOnlTra. Keeping both blocks prevents an intercept-only matrix calculation from being misread as global non-cancellation. The exact $H \mapsto 3H$ two-block transfer is `residueTransportDefect_three_eq_affineState` ExpOnlTra.

Every old channel $d \mid H$ is killed by the first-order defect `residueTransportDefect_eq_zero_of_dvd` ExpOnlTra. The endpoint `irrational_totient_series_of_exponentOnlyThreeTransportSupply` ExpOnlTra is conditional on a named fixed-three tower supply.

More generally, coefficients satisfying $\sum_i c_i = \sum_i c_i m_i = 0$ annihilate every old affine channel. This is `oldChannel_affine_moment_annihilation` JoiExpTra. For $q(X,Y) = XY - 3X - 2Y + 4$, the vanishing $q(1,1) = q(3,5) = 0$ gives

$$K_d(15H) - 3K_d(3H) - 2K_d(5H) + 4K_d(H) = 0 \quad (d \mid H),$$

checked by `joint35_oldChannel_zero` JoiExpTra. The associated window radius is $19H + 5L + 5$; the resulting finite non-integrality criterion is `joint35Cone_notMem_int_of_cert` JoiExpTra. The conditional endpoint is `irrational_totient_series_of_anchoredJoint35ConeSupply` JoiExpTra.

5 Prime jumps and three-transport boundaries

When $p \nmid H$, a jump from H to pH creates new squarefree support. The residue Hecke defect splits into a post-jump channel and an explicit migration correction in `residueHeckeDefect_eq_postJump_add_migration` PriJumMig. For $d \mid H$ the total old-channel defect is zero PriJumMig; the $(d, H, p) = (5, 12, 5)$ calculation records the two cancelling terms.

Set

$$J(H, p) = (R_{2pH} - R_{pH}) - p(R_{2H} - R_H).$$

Its exact remainder radius is $3pH + (p + 1)(L + 2)$, proved with the window split by `primeJumpTailCommutator_eq_window_add_remainder` PriJumWin. The finite non-integrality criterion is `primeJumpTailCommutator_notMem_int_of_sharpKill` PriJumWin. Lean verifies $(H, p, L) = (12, 5, 15)$ in `primeJumpSharpKill_twelve_five` PriJumWin; this is one finite deposit, not an unbounded sequence.

The balanced vertices $H, 2H, 3H, 6H$ give a related radius $9H + 4L + 4$. The corresponding criterion is `threeTransport_notMem_int_of_cert` ThrTraBou; the irrationality theorem `irrational_totientSeries_of_roughnessBoundaryLatticeSupply` ThrTraBou is conditional. The concrete check $(H, L) = (60, 12)$ is `sharpThreeTransportCert_60_12` ThrTraBou.

6 A first-harmonic criterion

For the window discrepancy $A(h, N, L)$ define

$$W_1(h, N, L) = \cos\left(2\pi \frac{A(h, N, L) \bmod 2^L}{2^L}\right).$$

Assume $N \in [X, 2X)$ and $16(2X + h + L + 2) \le 2^L$. If every N failed the central-residue certificate, each phase would lie within $\pi/8$ of an integer turn, so each cosine would exceed $9/10$. Hence

$$\sum_{X \le N < 2X} W_1(h, N, L) \le \frac{9}{10} X$$

forces a certificate in the block. This implication is `exists_certifiedKill_of_first_harmonic_gap` FirHarGap. More generally, if T is any nonempty finite set of indices below $2X$ and

$$\sum_{N \in T} W_1(h, N, L) \le \frac{9}{10} |T|,$$

then a certificate occurs at some $N \in T$. This subset form is `exists_certifiedKill_of_first_harmonic_gap_subset` FirHarGap. By certificate completeness, the witness produced here is an exact non-integral tail difference, not a heuristic phase test. Hence a first-harmonic saving on one finite block produces exactly the kind of witness required by the tail-difference criterion. What is still missing is a first-harmonic estimate that supplies such blocks at unbounded parameters; no such estimate is proved here.

7 Adjacent-pair phase separation

At a scale satisfying the same room inequality, with $N+1 < 2X$, suppose that the first harmonic phases at two adjacent indices obey

$$\|\text{windowFirstExp}(h, N + 1, L) - \text{windowFirstExp}(h, N, L)\|^2 \ge \frac{19}{25}.$$

Then a certificate occurs at one of N and $N + 1$. This finite implication is `exists_certifiedKill_of_phasePairSeparation` AdjPhaSep. Consequently, a cofinal supply of adjacent pairs satisfying this inequality implies irrationality of the totient constant, by `irrational_totient_series_of_adjacentPhaseSeparation` AdjPhaSep. No cofinal adjacent-pair separation is proved here.

8 Pivot anti-reconstruction and window-pair geometry

The first-harmonic phase of a finite window is the exact additive character of its dyadic discrepancy residue. It differs from the corresponding infinite tail-orbit phase by the explicitly bounded omitted tail. On a finite supplier fibre, the anchored defect is the squared distance of the window phases from the constant phase 1; its centred form is the usual pairwise-energy identity. Thus a counted family of separated phase pairs of sufficient total squared chord mass produces a finite certificate.

The formal proof records several equivalent sufficient cofinal hypotheses. In particular, a cofinal pivot anti-reconstruction bound, a cofinal centred reconstruction-variance bound, or a cofinal counted-separated-pairs bound is sufficient for irrationality of S. The direct finite criterion is `exists_certifiedKill_of_pivotFiberSeparatedPairs` PivAntRec, and the resulting irrationality theorem is `irrational_totient_series_of_pivotAntiReconstruction` PivAntRec.

There is also an exact fibre-free formulation: the cofinal window-pair predicate is equivalent to irrationality of S, not merely sufficient for it. The reverse implication uses the expansive doubling orbit to obtain adjacent infinite-tail separation and then chooses a finite window depth. This is formalised by `dtwWindowSeparatedPairs_iff_irrational_totient_series` PivAntRec. The development does not prove any of these cofinal arithmetic hypotheses, so it does not settle Erdős #249.

9 Fixed local precision cannot finish the argument

A finite valuation–unit symbol records a bounded amount of local transport data. For every positive precision u, every finite compatible word admits a prefix-locked centred completion. Thus no finite word over this fixed local alphabet excludes all centred endpoints. The exact statement is `fixedPrecisionTropicalNoGo` TroCurCar. This does not say curvature certificates are rare. A proof may use precision growing with scale, global correlations, or a harmonic gap. It rules out only the inference that a bounded local signature alone forces escape.

10 LCM factor ideals and whole-ray anchors

For every $t \geq 3$, there is a nonzero synthetic affine carry whose forcing letters lie in the principal ideal generated by $\varphi(\mathrm{lcm}(1, \dots, t))$, satisfy the exact whole-ray anchor values, and remain inside the relevant diagonal bounds. Nevertheless its state is a strict survivor. The construction is formalised by `lcm_factorIdeal_sparseAnchor_not_sufficient` LcmFacIdePulObs.

The same example remains a dyadic coboundary after every finite integer shift polynomial. In particular, the finite-rank LCM shift algebra alone cannot remove its terminal compensation; the corresponding formal statement is `lcm_factorIdeal_finiteRank_shiftAlgebra_not_sufficient` LcmFacIdePulObs. This is a synthetic countermodel: it does not assert that these forcing letters are actual totient differences, and it does not rule out a nonlinear argument using the missing fresh divisor channels.

11 Statement matrix and open boundary

Layer	Checked statement	Status
Curvature	exact split, radius, recurrence, criterion	conditional reduction
Exponent transport	stable support, affine state, cancellation	proved here
Joint $(3, 5)$ transport	moment annihilation and criterion	conditional reduction
Prime jump	migration, sharp window, $(12, 5, 15)$	verified finite instance
Three transport	sharp window and $(60, 12)$	verified finite instance
First harmonic	cosine gap forces a certificate	conditional reduction
Adjacent phases	squared chord separation forces a certificate	conditional reduction
Pivot anti-reconstruction	pairwise phase energy and exact window-pair equivalence	conditional reduction
Tropical signature	fixed precision has a completion	unconditional progress
LCM factor ideals	factor/anchor and finite shift algebra have a synthetic survivor	unconditional progress
Irrationality of S	follows from a named unbounded supply	open

The development proves no cofinal supply of curvature, prime-jump, joint, or three-transport certificates, and no cofinal first-harmonic gap. It therefore does not prove Erdős #249.

References

[1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de l'Enseignement Mathématique 28 (1980), 61–62, scan.

[2] T. Bloom (ed.), *Erdős Problems*, entry 249, `erdosproblems.com/249`, accessed July 2026.

[3] W. Cook, *Erdős Problems 249 and 257 in Lean 4: formalised results, certificate reductions, and open questions*, companion exposition in this release.